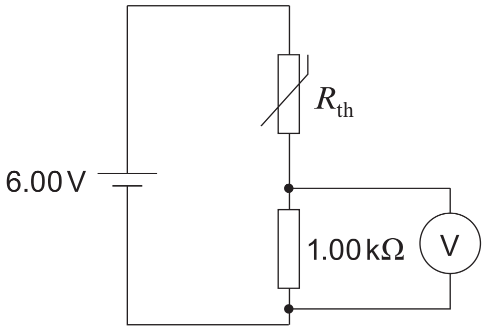
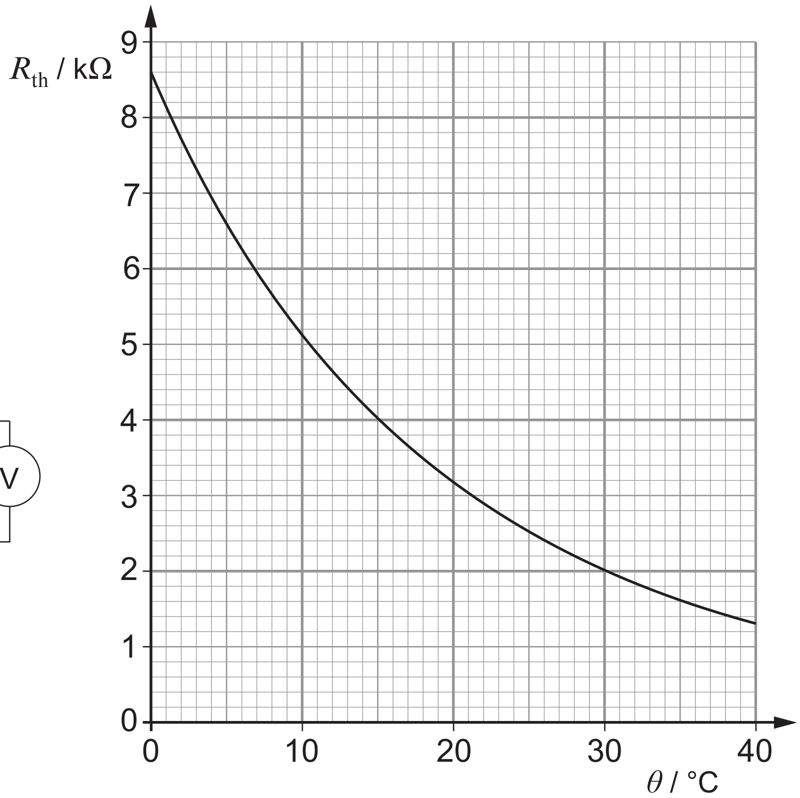
Examiner only

2. The resistance, R_{th}, of the thermistor in the circuit varies with temperature as shown in the graph.





(a) Determine the temperature of the thermistor when the voltmeter reads 1.20 V. [3]

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Examiner only

(b) The set-up can be used as a thermometer, with the thermistor used as a temperature probe. David suggests that there must be a simple rule of the type "an increase of 0.10 V in the voltmeter reading corresponds to a temperature increase of n°C in which n is a constant." Without further calculation, discuss whether he is right. [2]

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(c) Explain why the current through the thermistor must be very low, in order for it to work properly as a probe to measure the temperature of its surroundings. [2]

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